

TECHNICAL SKILLS

Languages: Python, C, Java, JavaScript, TypeScript, SQL, HTML/CSS, PHP**Databases:** PostgreSQL, Microsoft SQL Server, MongoDB, Firebase Firestore**Web Technologies:** React, Next.js, Node.js, Express, REST APIs, Tailwind CSS**Frameworks & Libraries:** Pandas, NumPy, Matplotlib, BeautifulSoup, Tkinter**Tools & Platforms:** Git/GitHub, PowerBI, Supabase, Vercel, AWS, Android Studio, Windows, Linux, Jupyter Notebook

EDUCATION

Bachelor of Science, Computing Science – Specialization

Sep. 2023 – Present

University of Alberta

Edmonton, AB

Relevant Coursework: Data Structures and Algorithms, Object-Oriented Programming, AI Search and Planning, Statistics

PROJECTS

CountLab.ca | TypeScript, Next.js, Tailwind CSS, Supabase, PostgreSQL, Python, Numba, Power BI

- Architected and deployed a full-stack, mobile-first blackjack training PWA with Supabase authentication, owner-scoped Row-Level Security, offline caching, and cross-device synchronization of user progress and session history.
- Reduced first-load JavaScript by **63%** (311 KB to 116 KB) and route-bundle size by **96%** (192 KB to 8.28 KB) through route-level code splitting, dynamic imports, memoization, and Web Worker isolation.
- Built Power BI user-analytics dashboards backed by a privacy-preserving Supabase/PostgreSQL pipeline instrumenting **84 strongly typed events**, 2–10-step conversion funnels, D1/D7/D30 retention cohorts, feature adoption, and Core Web Vitals.
- Developed a bankroll-analysis engine supporting Kelly-optimal bet ramps, lifetime and finite-trip risk of ruin, EV, variance, c-SCORE, DI, and N_0 calculations across **9 deck/penetration profiles**.
- Engineered a reproducible, parallelized Python Monte Carlo pipeline that simulated **116.8B+ blackjack rounds** across **250M shoes per profile**, producing per-true-count EV and variance estimates with **95% confidence intervals**.
- Implemented an exact game-theoretic solver using Numba-compiled backward induction and bitmask-based card representations, exhaustively evaluating **18B+ terminal assignments** for a single opening decision.

Prominent Person Stock Tracker | Python, Pandas, BeautifulSoup, Tkinter

- Built a desktop stock-tracking application visualizing publicly disclosed trades from executives and politicians across multiple data sources.
- Scraped and aggregated data using BeautifulSoup, processing millions of data points through a structured Pandas pipeline.
- Developed interactive filtering by individual and ticker, plus downloadable charts and CSV exports.

Android Social Media Mood App | Java, Firebase, Android Studio

- Developed a scalable Android application in an Agile team using GitHub workflows.
- Optimized database queries and asynchronous handling, improving efficiency by 74%.
- Designed a distributed Firebase backend with client-side caching.

Distributed Social Networking Platform | Python, Django, PostgreSQL, REST APIs, Heroku

- Collaborated in a 5-person team to design and test a federated social networking platform and validate cross-node interactions.
- Developed API test cases covering authors, posts, comments, likes, follow requests, authentication rules, edge cases, and pagination.
- Investigated defects by reproducing bugs, analyzing root causes, and verifying fixes before deployment.

Ray Tracing Engine | C

- Optimized 3D rendering algorithms, improving render time by 100%.
- Implemented advanced lighting and shading models.

EXPERIENCE

Dealer + Training Software Developer (GatewayTraining.ca)

Sep. 2024 – Present

Gateway Casinos and Entertainment

Edmonton, AB

- Architected a full-stack React, TypeScript, Node.js, and PostgreSQL platform with 112 REST endpoints backed by 17 relational tables and 13 database migrations.
- Reduced repeat loading across 10 frequently revisited views by implementing request deduplication, 5-minute response caching, background revalidation, and mutation-aware cache invalidation.
- Streamlined daily-report entry with automatically populated trainee rosters, 700-ms draft autosaving, integrated drill timers, quick score entry, and one-click report and timeline duplication.
- Built 7 analytics APIs and a centralized operations dashboard aggregating attendance, enrollment, payroll hours, class performance, missing reports, scheduling gaps, and recent activity.
- Exported to Power BI and presented to management team for reporting and analysis.
- Designed a PostgreSQL schema modeling Coordinator → Trainer → Student workflows with Supabase authentication, row-level security, and file storage for reports and PDFs.
- Built a React + TypeScript frontend enabling coordinators to manage classes and trainer assignments, trainers to create drills and grade students, and students to track performance metrics and drill times.